JAVASCRIPT QUIZ APP

Beginner-friendly project • HTML + CSS + JavaScript • Complete source code

Project overview

This interactive quiz application tests basic JavaScript knowledge through multiple-choice questions. It checks answers instantly, highlights correct choices, tracks the score, displays the final result, and lets users restart the quiz.

How to run the project

1. Open Notepad or a code editor such as VS Code.
2. Copy the complete source code from this PDF into a new file.
3. Save the file as **index.html**.
4. Double-click index.html to open the quiz in a web browser.
5. To customize it, edit the questions and answers inside the **questions** array.

Complete source code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>JavaScript Quiz App</title>
<style>
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #edf3fa; color: #1e293b; }
.container { max-width: 720px; margin: 55px auto; padding: 28px; }
.quiz { background: white; padding: 28px; border-radius: 16px; box-shadow: 0 8px 28px #1834511a; }
h1 { margin-top: 0; color: #245b8f; }
#question { font-size: 20px; font-weight: bold; margin: 22px 0; }
#answers { display: grid; gap: 10px; }
.answer { text-align: left; padding: 13px; border: 1px solid #ccd8e5; border-radius: 8px; background: #fff; cursor: pointer; font-size: 15px;
.answer:hover:not(:disabled) { background: #eef6ff; border-color: #245b8f; }
.answer:disabled { cursor: default; }
.answer.correct { background: #d9f4e3; border-color: #27864b; }
.answer.wrong { background: #ffe1df; border-color: #c8443b; }
.row { display: flex; justify-content: space-between; align-items: center; gap: 12px; margin-top: 20px; flex-wrap: wrap; }
button.primary { background: #245b8f; color: white; border: 0; border-radius: 8px; padding: 11px 18px; cursor: pointer; font-size: 15px; }
button.primary:disabled { opacity: .45; cursor: not-allowed; }
#feedback { min-height: 24px; font-weight: bold; margin-top: 12px; }
#result { font-size: 18px; font-weight: bold; }
@media (max-width: 600px) { .container { margin: 18px auto; padding: 12px; } .quiz { padding: 20px; } }
</style>
</head>
<body>
<main class="container">
  <div class="quiz">
    <h1>JavaScript Quiz</h1>
    <p>Choose the correct answer for each question.</p>
    <p id="progress">Question 1 of 5</p>
    <div id="question"></div>
    <div id="answers"></div>
    <p id="feedback" aria-live="polite"></p>
    <div class="row">
      <span id="score">Score: 0</span>
      <button class="primary" id="next" disabled>Next Question</button>
    </div>
    <div id="result" aria-live="polite"></div>
    <button class="primary" id="restart" hidden>Restart Quiz</button>
  </div>
```

```
</main>
<script>
const questions = [
  { q: "Which keyword declares a variable that can be reassigned?", a: ["const", "let", "fixed", "define"], correct: 1 },
  { q: "Which symbol is used for a single-line comment in JavaScript?", a: ["<!--", "//", "**", "##"], correct: 1 },
  { q: "Which method selects an element by its ID?", a: ["getElementById()", "getClass()", "selectID()", "findElement()"], correct: 0 },
  { q: "Which data type represents true or false?", a: ["String", "Number", "Boolean", "Array"], correct: 2 },
  { q: "Which event commonly runs when a button is clicked?", a: ["onhover", "onclick", "onloadtext", "onwrite"], correct: 1 }
];

let current = 0;
let score = 0;
let answered = false;

const questionEl = document.getElementById("question");
const answersEl = document.getElementById("answers");
const progressEl = document.getElementById("progress");
const scoreEl = document.getElementById("score");
const feedbackEl = document.getElementById("feedback");
const nextBtn = document.getElementById("next");
const resultEl = document.getElementById("result");
const restartBtn = document.getElementById("restart");

function showQuestion() {
  answered = false;
  const item = questions[current];
  progressEl.textContent = `Question ${current + 1} of ${questions.length}`;
  questionEl.textContent = item.q;
  answersEl.innerHTML = "";
  feedbackEl.textContent = "";
  resultEl.textContent = "";
  nextBtn.disabled = true;
  item.a.forEach((answer, index) => {
    const button = document.createElement("button");
    button.className = "answer";
    button.textContent = answer;
    button.addEventListener("click", () => chooseAnswer(index, button));
    answersEl.appendChild(button);
  });
}

function chooseAnswer(index, button) {
  if (answered) return;
```

```
  answered = true;
  const correctIndex = questions[current].correct;
  [...answersEl.children].forEach((el, i) => {
    el.disabled = true;
    if (i === correctIndex) el.classList.add("correct");
  });
  if (index === correctIndex) {
    score++;
    feedbackEl.textContent = "Correct answer!";
  } else {
    button.classList.add("wrong");
    feedbackEl.textContent = "Not quite. The correct answer is highlighted.";
  }
  scoreEl.textContent = `Score: ${score}`;
  nextBtn.disabled = false;
}

nextBtn.addEventListener("click", () => {
  current++;
  if (current < questions.length) {
    showQuestion();
  } else {
    questionEl.textContent = "Quiz completed!";
    answersEl.innerHTML = "";
    feedbackEl.textContent = "";
    progressEl.textContent = "Final result";
    resultEl.textContent = `You scored ${score} out of ${questions.length}.`;
    nextBtn.disabled = true;
    nextBtn.hidden = true;
    restartBtn.hidden = false;
  }
});

restartBtn.addEventListener("click", () => {
  current = 0;
  score = 0;
  scoreEl.textContent = "Score: 0";
  nextBtn.hidden = false;
  restartBtn.hidden = true;
  showQuestion();
});

showQuestion();
```

```
</script>
</body>
</html>
```

Features and concepts

• **HTML:** page structure, question area, answer buttons, score, and result display.

• **CSS:** responsive layout, answer states, buttons, spacing, and visual feedback.

• **JavaScript:** arrays, variables, functions, event listeners, conditions, and DOM updates.

• **Scoring:** correct answers increase the score and the final score is shown at the end.

• **Restart option:** resets the score and starts the quiz again.

Note: This is a front-end educational project. The quiz runs in the browser and does not require a database or internet connection after the file is saved.